

Voronoi Analysis of Categorical Data

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May 4, 2026

TBD

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1 Introduction

1.1 Homogeneity Analysis

Homogeneity analysis, a.k.a. *Multiple Correspondence Analysis*, can be introduced in many different ways. We refer to Guttman (1941), Tenenhaus and Young (1985), Bekker and De Leeuw (1988), Gifi (1990), Greenacre (1991), Le Roux and Rouanet (2010), De Leeuw (2023) for a bouquet of discussions and interpretations of the equations and algorithms defining the technique.

For our purposes a geometric interpretation using *star plots* seems the most natural one. We have n objects measured by m categorical variables. Variable j has k_j categories, which are the different values the variable can assume for the n objects. Suppose the objects are mapped into n points of Euclidean space $\mathbb{R}^p$. For each variable we can construct an additional k_j points in the same space by computing for each category the centroids of all object-points in that category. Thus each variable defines its own plot. We then draw lines from the n object points to “their” category point and we obtain k_j “stars”. Homogeneity analysis maps the objects into space in such a way that the stars are as small as possible, where the size of the star is measured as the sum of squares of the distances between each category points and “its” object points (summed over categories and over variables). In order to avoid the trivial solution in which all objects are mapped into the same point homogeneity requires that the *configuration* of object points (an $n \times p$ matrix) is column-centered and orthonormal.

There are a huge number of variations possible on the same theme. Instead of the centroids we could use the Weber points to define our stars. Instead of Euclidean distance we could use any other distance on $\mathbb{R}^p$. Instead of the stars we could use the convex hulls or the minimum spanning trees, or any other way to visualize and quantify the compactness of the point clouds of each category. Any one of these choices defines a technique that could be called homogeneity analysis, but in this paper we mean the classical choice that uses stars, squared Euclidean distance, and orthonormal object points.

The key mathematical object in homogeneity analysis is the *indicator matrix*, which is a classical way to code categorical data. For a variable j with k_j categories it is an $n \times k_j$ binary matrix G_j in which each row i contains a single element equal to one, indicating into which category the object i maps for variable j . The cross product $D_j := G_j' G_j$ is diagonal, with the *marginal frequencies* of the categories on the diagonal. For two different variables the crossproduct $C_{j\ell} = G_j' G_\ell$ is the $k_j \times k_\ell$ *cross table* or *contingency table* of the two variables.

The G_j can be used as weights in the traditional “stress” multidimensional scaling loss function

(Kruskal (1964a), Kruskal (1964b)), defined as

$$\sigma(X, Y_1, \dots, Y_m) := \sum_{j=1}^m \sum_{i=1}^n \sum_{\ell=1}^{k_j} g_{i\ell}^j (\delta_{i\ell}^j - d(x_i, y_\ell^j))^2. \quad (1)$$

Here the dissimilarity $\delta_{i\ell}^j = 0$ if $g_{i\ell}^j = 1$. If $g_{i\ell}^j = 0$ the dissimilarity is arbitrary, because that particular triple (i, j, ℓ) does not contribute to the loss function. Since $\delta_{i\ell}^j g_{i\ell}^j = 0$ for all (i, j, ℓ) we have

$$\sigma(X, Y_1, \dots, Y_m) = \sum_{j=1}^m \sum_{i=1}^n \sum_{\ell=1}^{k_j} g_{i\ell}^j \{d(x_i, y_\ell^j)\}^2. \quad (2)$$

Define

$$\sigma_\star(X) := \min_{Y_1, \dots, Y_m} \sigma(X; Y_1, \dots, Y_m). \quad (3)$$

The minimum over the Y_j is attained at the centroids $\hat{Y}_j := D_j^{-1} G_j' X$, and if we substitute these optimal $\hat{Y}_j$ in ... we see that $\sigma_\star(X)$ is indeed equal to our measure of the total size of all stars. Minimizing $\sigma_\star(X)$ over $X'X = I$ is homogeneity analysis, and computing the solution turns out to be a straightforward eigenvalue-eigenvector problem. The details are in the publications we mentioned before.

1.2 Voronoi Analysis

Although using loss function (1) makes homogeneity analysis a form of (metric) multidimensional scaling there are some important differences. There are no local minima in homogeneity analysis because it is a symmetric eigenvalue problem. Also, and most importantly, we do not really expect the minimum loss function value to be zero, or even close to zero. Zero loss would imply all object points coincide with all category points they map into, which is clearly unrealistic. It can only be realized if all object and category points map into a single point, which is prohibited by the constraint $X'X = I$. Thus we get a perfectly determinate and stable solution, but at the cost of imposing unrealistic constraints and an arbitrary normalization.

This makes it interesting to generalize loss function (1) to

$$\sigma(X, Y_1, \dots, Y_m, \Delta_1, \dots, \Delta_m) := \sum_{j=1}^m \sum_{i=1}^n \sum_{\ell=1}^{k_j} w_{i\ell}^j (\delta_{i\ell}^j - d(x_i, y_\ell^j))^2 \quad (4)$$

with fixed non-negative weights, and with the constraints that if $g_{i\ell}^j = 1$ then $\delta_{i\ell}^j \leq \delta_{i\nu}^j$ for all $\nu = 1, \dots, k_j$. Thus if object i is in category ℓ of variable j we do not require the object

point to coincide with the category point, but we merely require that the object point is at least as close to category point ℓ as to any of the other $k_j - 1$ category points of that variable.

The geometry corresponding with loss function (4) is different. As in homogeneity analysis there is a plot for each variable j with both n object points and k_j category points. The k_j category points define k_j *Voronoi regions* that partition the space. Voronoi region $\mathcal{V}_\ell^j$ is the region of space where the points are at least as close to category point y_ℓ^j as to any of the other category points. And loss (4) is zero (for a variable) if all object points are in the “correct” Voronoi region, i.e. in the Voronoi region of the category point corresponding with the category they are in.

One (very inefficient but conceptually easy) way of constructing Voronoi regions is to draw the perpendicular bisectors of the lines connecting all pairs of category points of a variable. The literature on properties and computation on Voronoi regions is large, and we refer to the textbooks of (?) and (?).

Since the category point itself is obviously in its own Voronoi region and since Voronoi regions are convex it follows that if we make stars by connecting category points with objects points as in homogeneity analysis then we actually require each star to be in its Voronoi region. Of course stars are no longer directly related to the value of the loss function.

Voronoi analysis of categorical data, as defined here, minimizes loss function (4) over X and over the Y_j and over the Δ_j satisfying the ordinal constraints. It is consequently a form of non-metric (ordinal) multidimensional scaling. The constraint $X'X = I$ from homogeneity analysis is no longer needed, but some form of normalization is still necessary. We still need to exclude the trivial solution in which all points and regions collapse into a single point and all Δ_j are zero. But, more urgently, we have to deal with the trivial solutions familiar from multidimensional unfolding (Heiser (1981), De Leeuw (1983), Busing (2010)).

2 Algorithm

$$\sigma(X, Y, \Delta) = \sum_{i=1}^n \sum_{j=1}^m \|\delta_{ij} - d_{ij}(X, Y)\|^2$$

If $g_{i\ell}^j = 1$ then $\{\delta_{ij}\}_\ell \leq \{\delta_{ij}\}_\nu$ for all $\nu = 1, \dots, k_j$.

$$\delta'_{ij} J \delta_{ij} = 1$$

Majorization

$$A_{ij} := \begin{bmatrix} e_i e_i' & -e_i e_j' \\ -e_j e_i' & e_j e_j' \end{bmatrix}$$

$$Z := \begin{bmatrix} X \\ Y \end{bmatrix}$$

$$d_{ij}(X, Y) = \text{tr } Z' A_{ij} Z$$

$$V := \sum_{i=1}^n \sum_{j=1}^m A_{ij} = \begin{bmatrix} mI & -ee' \\ -ee' & nI \end{bmatrix}$$

If we define

$$V_1 := (n+m)^{-1} \begin{bmatrix} \frac{m}{n}E & -E \\ -E & \frac{n}{m}E \end{bmatrix}, \quad (5)$$

$$V_2 := \begin{bmatrix} J_n & 0 \\ 0 & 0 \end{bmatrix}, \quad (6)$$

$$V_3 := \begin{bmatrix} 0 & 0 \\ 0 & J_m \end{bmatrix}, \quad (7)$$

then

$$V = (n+m)V_1 + mV_2 + nV_3$$

Matrices V_1 , V_2 , and V_3 are symmetric projectors on orthogonal subspaces. It follows that V^+ , the Moore-Penrose inverse³ of V , is

$$V^+ = \frac{1}{n+m}V_1 + \frac{1}{m}V_2 + \frac{1}{n}V_3$$

$$B(X, Y) := \begin{bmatrix} B_{11}(X, Y) & B_{12}(X, Y) \\ B_{21}(X, Y) & B_{22}(X, Y) \end{bmatrix}$$

Now $V_1 B(X, Y) = 0$ and thus

$$V^+ B(X, Y) = \frac{1}{m}V_2 B(X, Y) + \frac{1}{n}V_3 B(X, Y)$$

3 The Cone

Consider the cone of all vectors x in $\mathbb{R}^n$ for which $x_i \leq x_1$ for all $i \neq 1$ and $e'x = 0$. The extreme rays of the cone will be the $n - 1$ vectors of the form $\alpha e_i + \beta(e - e_i)$ with $i \neq 1$. Thus element i is equal to α and all other elements (including x_1) are equal to β . Thus we must have $\alpha > \beta$. The sum of the elements is $\alpha + (n - 1)\beta$, which must be zero. Thus $\alpha = -(n - 1)\beta$, which implies $\beta < 0$ and $\alpha > 0$. The extreme rays are positive multiples of the vectors $e_i - n^{-1}e$ with $i \neq 1$ and the cone is the set of all weighted sums with non-negative weights of these $n - 1$ vectors. Here is a small example.

```
a <- -cbind(1, -diag(4))
b <- (diag(5)-(1 / 5))[, -1]
print(a)
```

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	-1	1	0	0	0
[2,]	-1	0	1	0	0
[3,]	-1	0	0	1	0
[4,]	-1	0	0	0	1

```
print(b)
```

	[,1]	[,2]	[,3]	[,4]
[1,]	-0.2	-0.2	-0.2	-0.2
[2,]	0.8	-0.2	-0.2	-0.2
[3,]	-0.2	0.8	-0.2	-0.2
[4,]	-0.2	-0.2	0.8	-0.2
[5,]	-0.2	-0.2	-0.2	0.8

```
print(a %*% b)
```

	[,1]	[,2]	[,3]	[,4]
[1,]	1	0	0	0
[2,]	0	1	0	0
[3,]	0	0	1	0
[4,]	0	0	0	1

For later reference the sum of squares of the vectors $e_i - n^{-1}e$ is $(n - 1)/n$, so the vectors $\sqrt{n/(n - 1)}(e_i - n^{-1}e)$ are the vectors of length 1 on the extreme rays. The average of the extreme rays is

4 Normalized Monotone Regression

Suppose y is a vector with n elements. The problem we study is to minimise the sum of squares $\sigma(x) := \|x - y\|^2$ over $x \in K$ and $x'Jx = 1$. Here

- K is the cone of vectors with $x_1 \leq x_2 \leq \dots \leq x_n$,
- J is the centering matrix $J := I - n^{-1}ee'$, where
- e is the vector with all n elements equal to one.

Change variables to $\tilde{x} = Jx$. Then $x'Jx = \tilde{x}'\tilde{x}$ and $x \in K$ if and only if $\tilde{x} \in K$. Also $\tilde{x} = Jx$ if and only if $x = \tilde{x} + \beta e$. So the transformed problem is to minimize $\|\tilde{x} + \beta e - y\|^2$ over β and $\tilde{x} \in K$, with $e'\tilde{x} = 0$ and $\tilde{x}'\tilde{x} = 1$. The minimizer for β is $\bar{y} := n^{-1}e'y$, and thus the problem becomes minimizing $\|\tilde{x} - Jy\|^2$ over $\tilde{x} \in K$, $\tilde{x}'\tilde{x} = 1$, and $e'\tilde{x} = 0$.

Forget about the constraint $e'\tilde{x} = 0$. If M is the monotone regression operator it follows that the solution for $\tilde{x}$ is $M(Jy)/\|M(Jy)\|$, which automatically satisfies $e'\tilde{x} = 0$. Thus the solution of our original problem is

$$\hat{x} = \frac{M(y - \bar{y})}{\|M(y - \bar{y})\|} + \bar{y}.$$

$M(y - \bar{y}) = M(y) - \bar{y}e$ and thus $\|M(y - \bar{y})\|^2 = \|M(y)\|^2 + n\bar{y}^2 - 2\bar{y}e'M(y)$ nor $e'M(y) = e'y = n\bar{y}$. Thus $\|M(y - \bar{y})\|^2 = \|M(y)\|^2 - n\bar{y}^2$.

$$\hat{x} = \frac{M(y) - \bar{y}e}{\|M(y) - \bar{y}e\|} + \bar{y}e = \frac{1}{\|M(y) - \bar{y}e\|} M(y) + \left\{ 1 - \frac{1}{\|M(y) - \bar{y}e\|} \right\} \bar{y}e$$

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